

EEG Sleep Classification and Reference-Based Sleep-Health Profiling

Biomarker-assisted N1/N2 refinement

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Abstract

BACKGROUND

Sleep-stage sequences describe sleep continuity and architecture. N1/N2 confusion can distort these profiles.

RESULTS

On the same 3,109 test epochs, accuracy rose from 55.45% to 59.34%; macro-F1 rose from 57.42% to 59.51%.

METHOD

Four-channel EEG is scored in 30-second epochs. A retained hybrid model is refined using nine spectral features and logistic regression.

FUTURE WORK

Develop coarse and NREM specialist models, combine their probabilities, add temporal context, and validate on external recordings.

Why detect sleep stages? Why this project?

Sleep supports memory, learning and physical health.

Duration alone does not show when sleep is interrupted or how stages are distributed.

What stage detection gives us

Sleep onset and Wake time → continuity
N1 / N2 / N3 / REM time → architecture
A hypnogram → when patterns change

Why we chose this project

EEG records brain activity used in staging.
HMC provides labels to test our predictions.
N1/N2 confusion is a measurable target.

Potential benefit: assist sleep-study review and quantify patterns for research.

Our contribution: targeted refinement + descriptive profiles. Stage labels alone do not diagnose disease.

Sleep stages and what a sleep cycle means

Stage	What it means	Why we measure it
Wake	Awake before or between sleep periods	Sleep onset and interruptions
N1	Transition from Wake to light sleep	Onset of sleep; distinguish N1 from N2
N2	Light non-REM sleep; brain activity slows	Time in N2; our refinement target
N3	Deep, slow-wave sleep; supports restoration	Time spent in deep sleep
REM	Active brain, rapid eye movements; dreaming is common	Timing and duration of REM sleep

A cycle combines NREM and REM; these stages recur across the night.

Typically, more N3 occurs early and REM periods lengthen later. Both phases support learning and memory.

Literature review and project direction

Study	Main contribution	Relevance to our project
DeepSleepNet Supratak et al., 2017	CNN features + BiLSTM across EEG epochs	Compare a learned sequence-context baseline
SeqSleepNet Phan et al., 2019	Hierarchical recurrent sequence-to-sequence staging	Add context between epochs; current model is within-epoch
U-Sleep Perslev et al., 2021	Fully convolutional staging across clinical cohorts	Test transfer and robustness on a separate dataset
RobustSleepNet Guillot & Thorey, 2021	Montage flexibility and leave-one-dataset-out testing	Check channel compatibility and external generalization
SLEEPYLAND Dei Rossi et al., 2025*	Standardized evaluation; SOMNUS soft-voting ensemble	Test model fusion and calibrate the final output

Design direction: targeted experts + sequence context + fair external evaluation

*2025 arXiv preprint. Published scores use different datasets and protocols; no direct ranking is claimed.

Dataset and 30-second EEG epochs

HMC v1.1: 151 clinical PSG recordings; project subset: 24 recordings

Partition	Recordings	EEG epochs
Training	16	15,216
Validation	4	3,619
Test	4	3,109
Total	24	21,944

One EEG epoch = one 30 s segment

0–30 s

30–60 s

60–90 s

At 100 Hz: $30 \times 100 = 3,000$ samples
Four channels → input shape $4 \times 3,000$

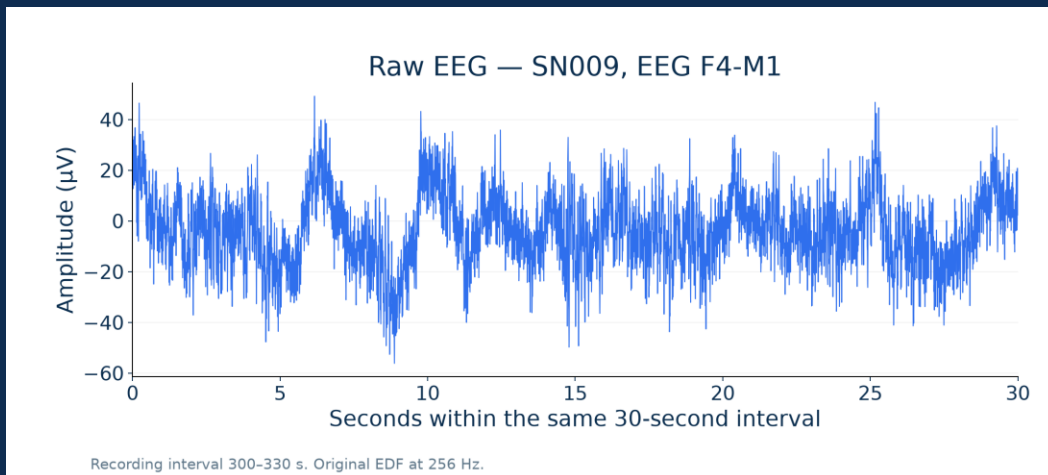
F4-M1 · C4-M1 · O2-M1 · C3-M2

Split by recording ID. SN001 covers only 190 epochs (95 min); whole-night claims are unsupported.

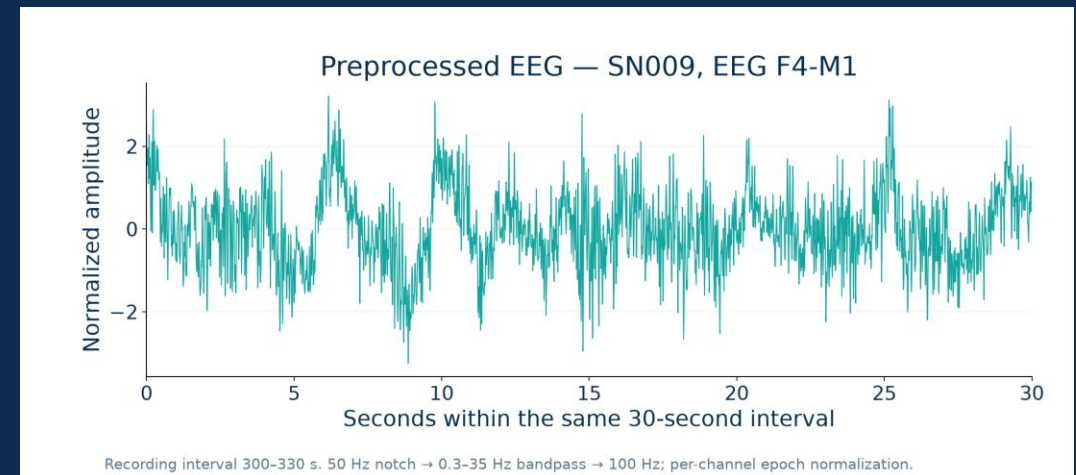
EEG preprocessing



Raw EEG · SN009, F4-M1



Filtered and normalized EEG · same interval

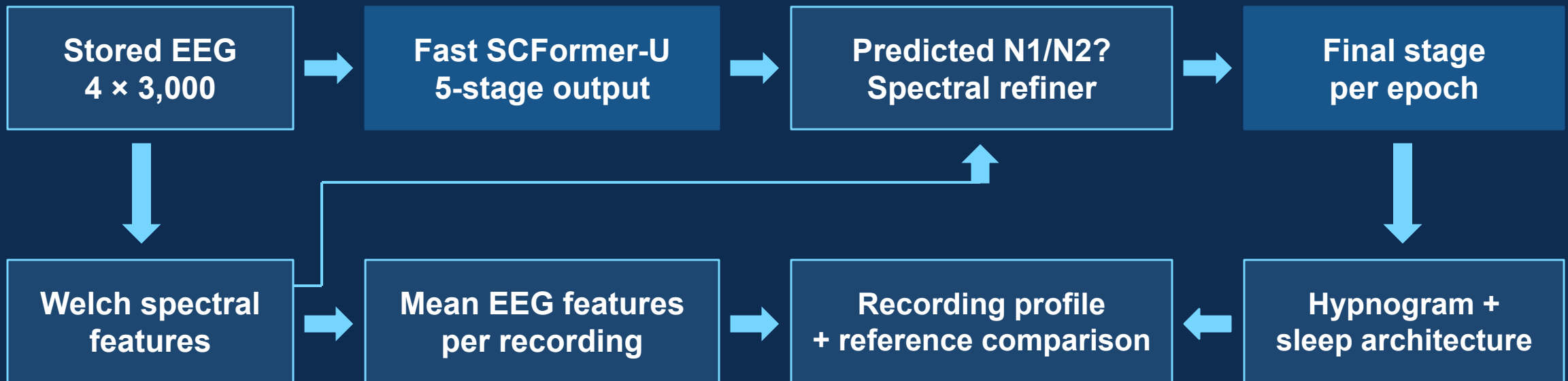


Per-epoch, per-channel normalization. Apply the 50 Hz notch before resampling.

Archived 300–330 s signal illustration only; all performance results use the current 3,109-epoch run.

Implemented system block diagram

Main implementation: Biomarker_N1_N2_Refinement.ipynb



Architecture + spectral features → z-scores / P10–P90 → four-domain profile → dashboard

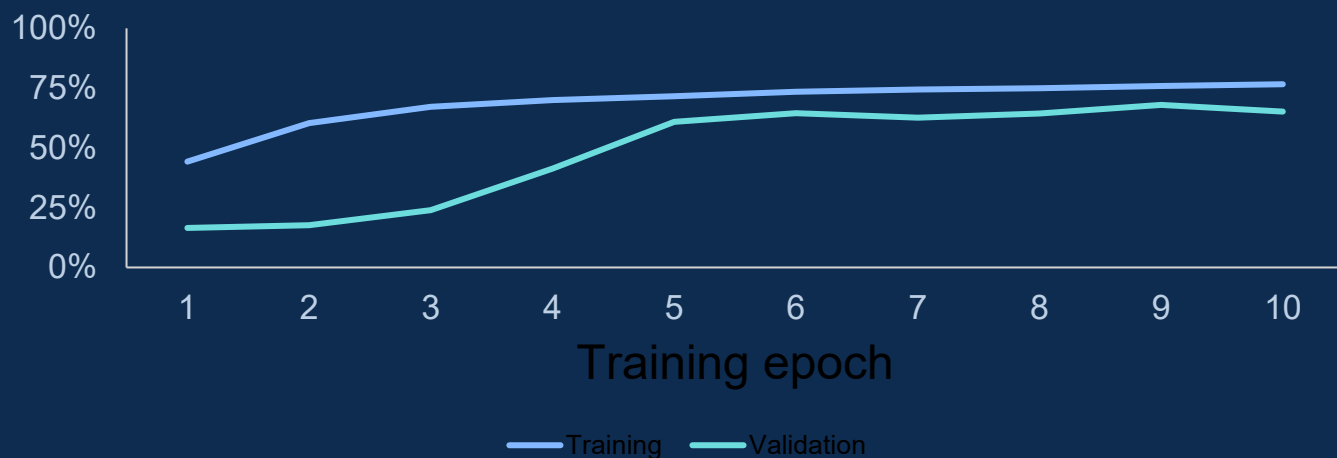
The 16-recording training reference provides dataset-relative descriptions.

Existing model and training epochs



226,309 parameters · Attention and BiLSTM operate within each 30-second EEG epoch

Training / validation accuracy



A training epoch = one full pass through training data

Adam · learning rate 0.0001

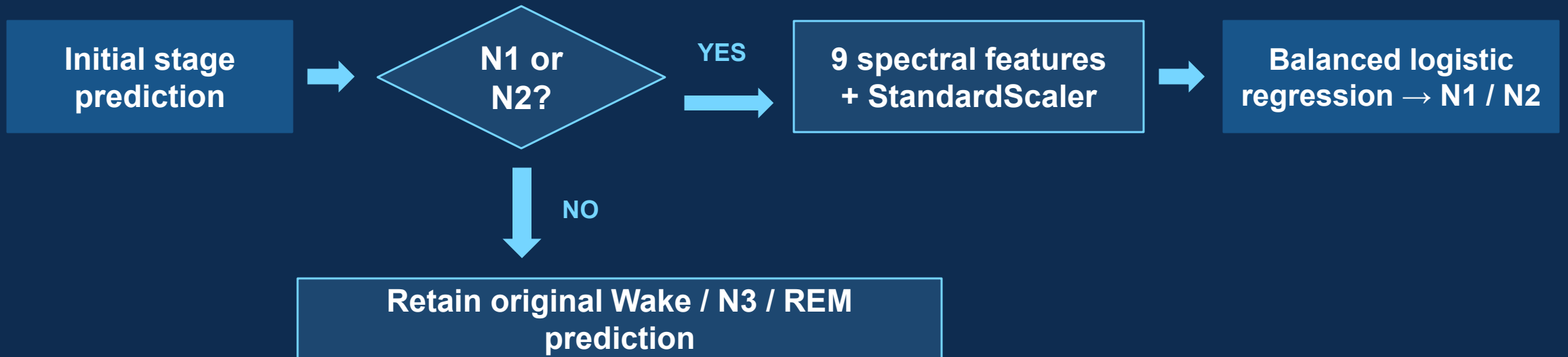
Batch size: 128

10 training epochs

Selected checkpoint: epoch 9

Current notebook loads this checkpoint; it trains the spectral refiner separately.

N1/N2 refinement flowchart



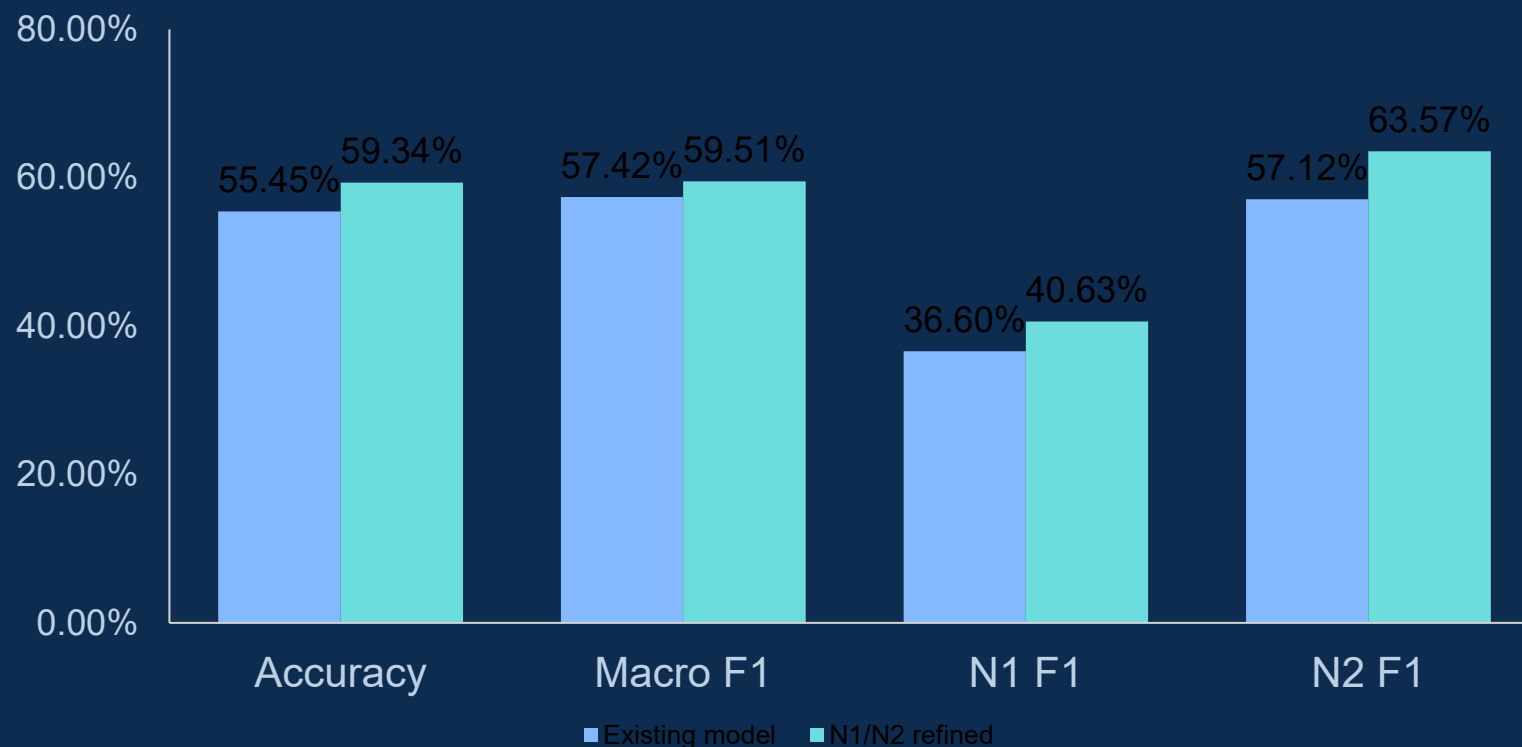
Feature group	Input to the refiner
Relative powers	Delta, Theta, Alpha, Sigma, Beta
Band / Delta ratios	Theta, Alpha, Sigma, Beta

Fit on training data only

True AND predicted N1/N2:
1,278 N1 + 4,588 N2 epochs
Test gate uses predictions only.

Welch: 4 s windows, 2 s overlap; average channel powers before computing ratios.

Results and measured gains



+3.89 pp

Accuracy: 55.45% → 59.34%

+2.10 pp

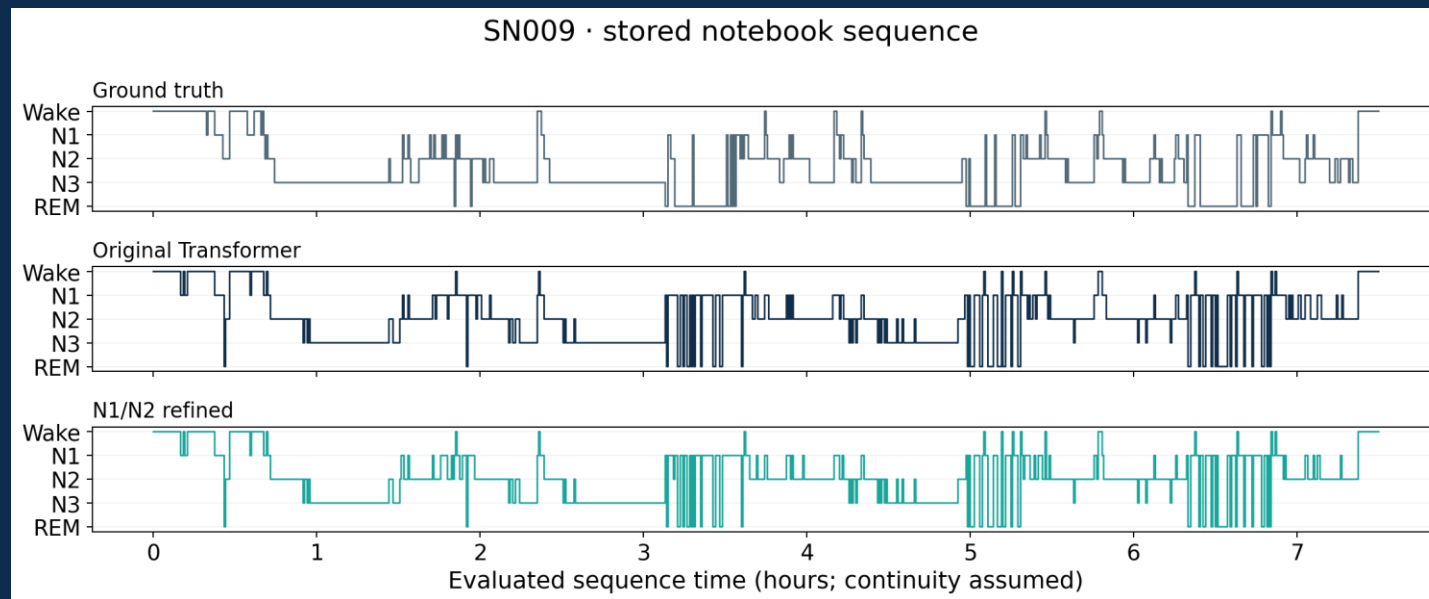
Macro F1:
57.42% → 59.51%

154 corrected – 33 degraded = 121 net additional correct epochs

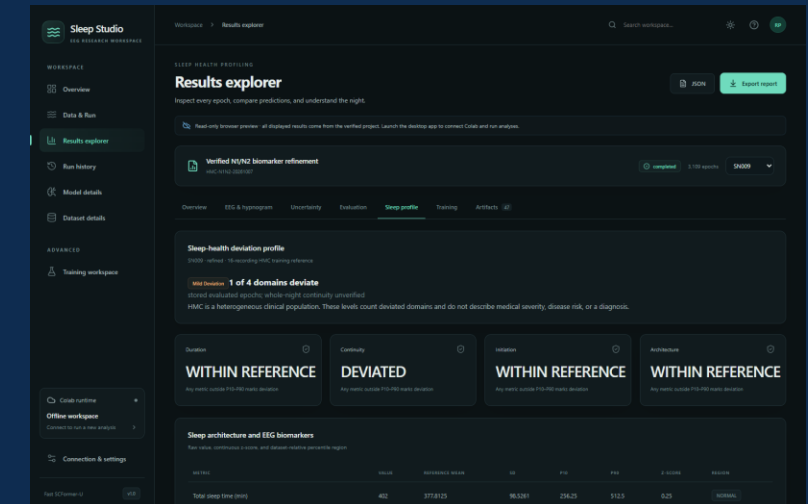
Same 3,109 test epochs across 4 recordings · 238 labels changed · pp = percentage points

Project outputs: hypnogram and sleep profile

SN009 · verified stage sequence



Sleep Studio · current GUI



Architecture, spectral features,
reference deviations and exports

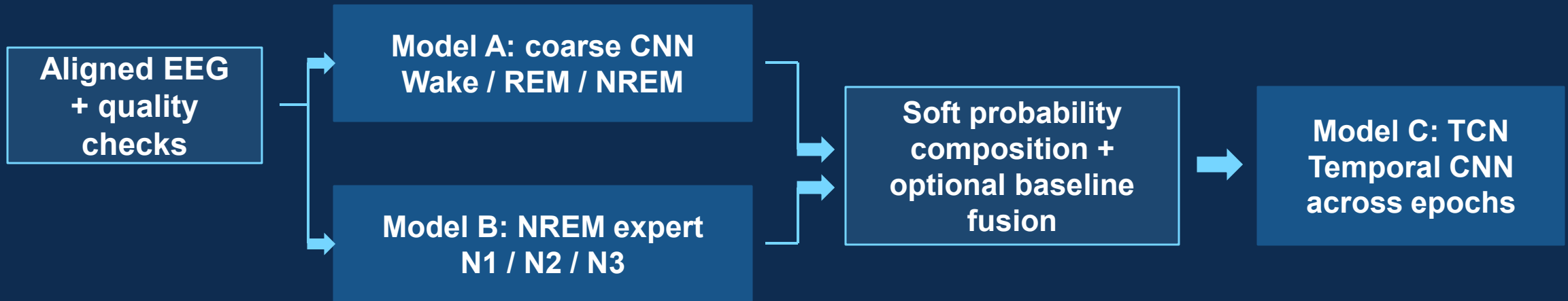
Stage labels → architecture metrics → training-reference comparison → four-domain profile

Duration · Continuity · Initiation · Architecture | Descriptive research profiles, not diagnoses

Notebook + verified run HMC-N1N2-20261007

Future plan: multistage, multimodel architecture

Proposed • not yet implemented or evaluated



Conditional NREM probabilities preserve five-class normalization

1 Experts

Test coarse + NREM models against the current baseline.

2 Context + confidence

Train TCN on out-of-fold outputs. Calibrate final probabilities.

3 Broader validation

New locked cohort, then external data and optional EOG / EMG.

Conclusion and next milestones

What is demonstrated

Spectral features improve N1/N2 discrimination in the current experiment.

59.34% test accuracy; 59.51% macro F1.
A complete notebook-to-GUI pipeline produces stage labels and profiles.

Limits: 4 test recordings, partial SN001,
N1 F1 40.63%, no clinical validation.

Next milestones

- 01 Audit full EDF alignment and quality; reserve a new locked test cohort.
- 02 Compare experts, soft fusion and TCN through controlled ablations.
- 03 Evaluate calibration, external data and biomarker errors; add modalities.

Evaluate macro F1, N1/N2 F1, calibration, latency and errors in sleep biomarkers.

Team members and project mentor

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